

Building the Glass Box: A Human-Centered Framework for Explainable AI in Cyber-Physical Systems

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May 2026

Presentation Outline

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The Rise of AI in Cyber-Physical Systems (CPS)

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CPS represent the profound integration of computation, networking, and physical processes. They form the backbone of modern critical infrastructure (e.g., Smart Power Grids, Autonomous Water Treatment Plants).

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The Catastrophic Vulnerability: The "Black Box"

Despite high accuracy, AI models act as opaque "black boxes". In safety-critical environments, when an AI flags an anomaly, human operators cannot answer: **"Why did the AI make this decision?"**

Literature Survey

Reference (IEEE Format)	Core Focus	Identified Gap / Limitation
Wickramasinghe et al. (2021) [1]	Explainable unsupervised ML for cyber-physical systems.	Context-Agnostic: Evaluates sensors in a vacuum without external environmental factors.
Ahmed et al. (2022) [2]	XAI applications in Industry 4.0 and smart manufacturing.	Evaluation Gap: Focuses on algorithmic accuracy; lacks formalized, human-centered evaluation standards.
Islam et al. (2021) [3]	Comprehensive survey of XAI approaches and metrics.	Metric Flaw: Evaluation metrics do not measure operational "Actionability" or "Trust" in high-stakes CPS.
Sofianidis et al. (2021) [4]	Review of ethical and human-centric AI in smart production.	Implementation: Purely theoretical framework; lacks empirical, multi-agent software implementation.

The Crisis of Traditional Explainable AI (XAI)

To mitigate the "black box" crisis, Explainable AI (XAI) frameworks like SHAP and LIME have become industry standards. However, they suffer from a severe operational limitation.

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Operational Consequence

Context-agnostic explanations generate costly false positives, misdiagnose environmental anomalies as physical hardware failures, and rapidly degrade operator trust.

Project Objectives & The "Glass Box" Approach

Primary Goal: To transition from theoretical XAI models to a functional, enterprise-ready software prototype that actively understands its environment.

- ① **Context-Aware Reasoning Engine:** Correlate internal mechanical/electrical telemetry with dynamic external variables (weather, network latency) to eliminate false positives.

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- ➍ **Automate Human-Centered Evaluation:** Pioneer an "LLM-as-a-Judge" pipeline to objectively score explanations, eliminating the bottleneck of manual human testing.

Redefining Metrics: Why Trust, Reasonableness, Actionability?

The Problem with Standard Metrics

Metrics like Accuracy or F1-scores measure if a model is *correct*, but not if the explanation is *useful*. Traditional XAI (like SHAP) provides numerical feature weights, not human-readable reasoning.

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- **Reasonableness ("Does this make logical sense?")**: Checks if the AI's logic aligns with established domain physics and expert engineering knowledge.
- **Actionability ("What do I actually do now?")**: An explanation is useless if it doesn't guide a concrete next step under time pressure.

The Data Challenge in Safety-Critical CPS

The Invigilator Question: Why not use real-world data?

While real-world telemetry is the gold standard for testing AI, accessing raw, labeled data from modern Cyber-Physical Systems is practically impossible for two critical reasons:

- **National Security & Air-Gapping:** Smart Power Grids and Water Treatment Plants are critical infrastructure. Their SCADA network telemetry is strictly proprietary, classified, and air-gapped to prevent cyber-terrorism.

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Our Solution: Synthesize a physics-grounded, dual-domain dataset that intentionally isolates mechanical failures from environmental anomalies to rigorously stress-test the XAI's reasoning.

Dataset Description & Source

To ensure high operational realism, we engineered two 1,000-sample time-series datasets (15-minute intervals) using Gaussian distributions centered on real engineering standards.

Smart Power Grid

- **Internal Base:** 132kV Voltage, 400A Current, 50Hz Frequency.
- **Contextual Anomalies:** Heatwaves causing thermal stress; Lightning causing latency.
- **Mechanical Failures:** Isolated Transformer/Line faults.

Smart Water Plant

- **Internal Base:** 50 psi Pressure, 2.0 mm/s Pump Vibration.
- **Contextual Anomalies:** Heavy Storms causing severe network routing disruption.
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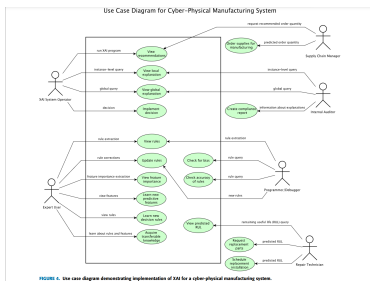
Ensuring Legitimacy: Our Python generation script binds all normal states within strict tolerances (e.g., Grid frequency $\approx 49.9 - 50.1$ Hz). When anomalies are injected, the telemetry mathematically deviates in ways that accurately mimic physical engineering laws.

Operational Workflow: XAI-CPS Ecosystem

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- **Persona-Driven Explainability:** An Operator needs immediate actionability, while an Auditor needs global rule extraction. Our framework is designed to serve this diverse ecosystem.

Data Preprocessing & Anomaly Filtering

Before the raw dataset is passed to our XAI Multi-Agent pipeline, it must undergo strict preprocessing and rule-based threshold filtering.

Preprocessing Steps:

- 1 **Data Standardization:** Raw telemetry is synchronized into 15-minute intervals to align internal mechanical sensors with external environmental data (weather, network latency).

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- 3 **LLM Triggering:** Computationally heavy LLM inference is completely paused until this preprocessing step flags a critical event, ensuring real-time system efficiency.

System Architecture: Step 2 - Multi-Agent Orchestration

Tech Stack: Microsoft AutoGen + LLaMA 3.2 (via Ollama, 100% Offline).

Once an anomaly is flagged, it is passed to a three-agent ecosystem:

- **Agent 1a (Context-Agnostic Explainer):** Represents traditional XAI. Receives *only* internal sensor data. Assumes mechanical failure by default.

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Note: By making the causal reasoning process explicit and transparent via LLM prompting, we move beyond "black-box" scores to true, human-understandable AI.

Empirical Results: The "Glass Box" Dashboard

We developed a functional human-machine interface using **Streamlit**, providing a central dashboard for CPS operators with a dynamic domain selector (Water vs. Power).

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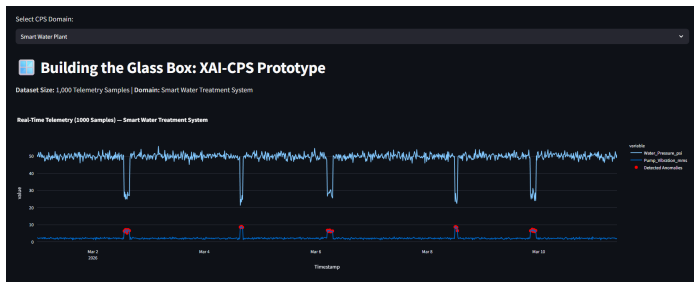


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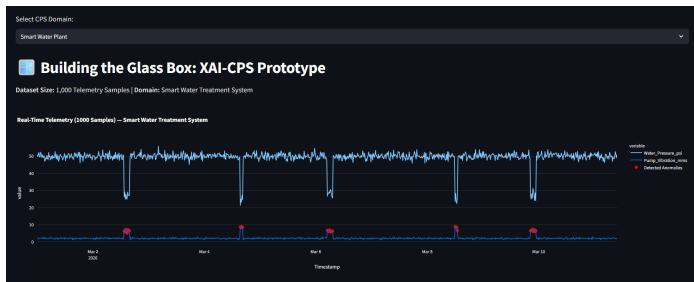


Figure: Real-time CPS sensor telemetry tracking detected anomalies.

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Comparative AI Diagnostics in Action

When an anomaly triggers the system, the dashboard explicitly contrasts the two explanation paradigms side-by-side:

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Analysis Results

Traditional XAI (Context-Agnostic)

AGENT EXPLANATION

- Anomaly Analysis:** An unusually high water pressure reading (25.80 psi) was detected at a pump vibration level that is significantly higher than normal (6.50 mm/s), indicating potential mechanical failure.
- Diagnosis:** The anomaly is likely caused by a mechanical failure within the pump system, as evidenced by the elevated pump vibration data and corresponding high water pressure reading.
- Impact:** This anomaly may cause the pump to fail or experience reduced performance, leading to disruptions in water supply and potentially affecting downstream systems.
- Recommended Action:** Investigate the pump system for signs of mechanical stress or damage, such as excessive wear on moving parts or unusual noise patterns, and perform a thorough inspection to identify the root cause of the anomaly.

EXPERT EVALUATION (SIMULATED)

Proposed Framework (Context-Aware)

AGENT EXPLANATION

- Anomaly Analysis:** Anomalous water pressure reading detected at 2024-03-08 13:00:00, indicating potential issues with the treatment system's hydraulic balance.
- Contextual Diagnosis:** The clear external context suggests that environmental factors are unlikely to be contributing to the anomaly. However, the high pump vibration rate of 6.50 mm/s may indicate a mechanical issue within the pumping system, which could be related to the water pressure reading.
- Contributing Factors:** Network latency of 28.81 ms is relatively low, suggesting that communication issues with the network are unlikely to be the primary cause of the anomaly. However, it's possible that sensor data transmission delays may have masked or exacerbated the issue.
- Impact Assessment:** The anomalous water pressure reading could lead to reduced treatment system efficiency, potentially affecting water quality and treatment effectiveness.
- Recommended Action:** Perform a visual inspection of the pumping system and check for any signs of mechanical wear or damage, while also reviewing historical sensor data to identify potential patterns or correlations with similar anomalies.

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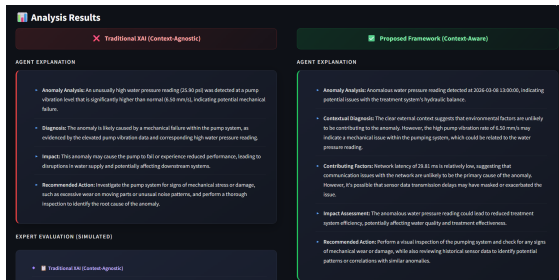
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- **Traditional XAI (Left):** Diagnoses a broken pump/transformer purely based on pressure/voltage drops. Recommends costly physical repair.

Comparative AI Diagnostics in Action

When an anomaly triggers the system, the dashboard explicitly contrasts the two explanation paradigms side-by-side:



- **Traditional XAI (Left):** Diagnoses a broken pump/transformer purely based on pressure/voltage drops. Recommends costly physical repair.
- **Proposed Framework (Right):** Ingests external variables to realize the grid is overworking to compensate for a heatwave/storm.

Result Discussion: Automated Expert Evaluation

Immediately following the explanations, the **Expert Evaluator Agent** outputs unbiased scorecards to assist the operator.

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EXPERT EVALUATION (SIMULATED)

- Traditional XAI (Context-Agnostic)
- Trust: 3/5 — The explanation is based on internal sensor data from the pump system, which may not provide a complete picture of the anomaly's causes or context.
- Reasonableness: 4/5 — While the analysis identifies potential mechanical failure as the likely cause, it does not consider external factors that could influence the anomaly, such as environmental conditions or network latency.
- Actionability: 4/5 — The recommended action is specific and actionable, requiring a thorough inspection to identify the root cause of the anomaly.

EXPERT EVALUATION (SIMULATED)

- Proposed Framework (Context-Aware)
- Trust: 4/5 — The explanation effectively uses contextual information to support its analysis, but it could benefit from more explicit quantification of confidence levels or uncertainty assessments.
- Reasonableness: 4/5 — The proposed diagnosis and contributing factors are reasonable, but the explanation could be strengthened by providing more specific details about the hydraulic balance issues and their potential causes.
- Actionability: 3.5/5 — While the recommended action is clear, it may benefit from additional guidance on how to prioritize or sequence the inspection and data review tasks, as well as potential mitigation strategies for any identified mechanical issues.

- Recommended Action: Perform a visual inspection of the pumping system and check for any signs of mechanical wear or damage, while also reviewing historical sensor data to identify potential patterns or correlations with similar anomalies.

Result Discussion: Automated Expert Evaluation

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The image displays two side-by-side panels, each titled "EXPERT EVALUATION (SIMULATED)".

Left Panel (Traditional XAI (Context-Agnostic)):

- Trust: 3/5 — The explanation is based on internal sensor data from the pump system, which may not provide a complete picture of the anomaly's causes or context.
- Reasonableness: 4/5 — While the analysis identifies potential mechanical failure as the likely cause, it does not consider external factors that could influence the anomaly, such as environmental conditions or network latency.
- Actionability: 4/5 — The recommended action is specific and actionable, requiring a thorough inspection to identify the root cause of the anomaly.

Right Panel (Proposed Framework (Context-Aware)):

- Trust: 4/5 — The explanation effectively uses contextual information to support its analysis, but it could benefit from more explicit quantification of confidence levels or uncertainty assessments.
- Reasonableness: 4/5 — The proposed diagnosis and contributing factors are reasonable, but the explanation could be strengthened by providing more specific details about the hydraulic balance issues and their potential causes.
- Actionability: 3.5/5 — While the recommended action is clear, it may benefit from additional guidance on how to prioritize or sequence the inspection and data review tasks, as well as potential mitigation strategies for any identified mechanical issues.

Recommended Action (from right panel): Perform a visual inspection of the pumping system and check for any signs of mechanical wear or damage, while also reviewing historical sensor data to identify potential patterns or correlations with similar anomalies.

- **The Result:** Our proposed context-aware model consistently achieves better scores in actionability, Reasonableness, and Trust by eliminating context-agnostic false positives.

Conclusion & Future Research Directions

Project Conclusion

We successfully transitioned from a theoretical methodology to a fully functional, 100% offline multi-agent XAI prototype. By proving cross-domain scalability and pioneering an LLM-as-a-Judge evaluation architecture, we demonstrated that context-aware, trustworthy AI is practically achievable for Industry 5.0.

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- ➋ **Multisensory Integration:** Translating AI diagnostics into haptic feedback (smart-gloves) or directional audio alerts to reduce cognitive load in noisy environments.

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We successfully transitioned from a theoretical methodology to a fully functional, 100% offline multi-agent XAI prototype. By proving cross-domain scalability and pioneering an LLM-as-a-Judge evaluation architecture, we demonstrated that context-aware, trustworthy AI is practically achievable for Industry 5.0.

Future Research Directions:

- ➊ **Expanding the Multi-Agent Ecosystem:** Adding specialized agents (e.g., "Autonomous Safety Auditor") to autonomously debate solutions before notifying the operator.
- ➋ **Multisensory Integration:** Translating AI diagnostics into haptic feedback (smart-gloves) or directional audio alerts to reduce cognitive load in noisy environments.
- ➌ **Biometric Validation:** Validating our automated LLM-as-a-Judge scores against human eye-tracking or EEG scans.

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Thank You

Questions & Discussion